EFY TESTED
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A Simple And Secure Way
TO SEND FREQUENT DATA ANYWHERE

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Sometimes, we need frequent sensor data transmission from one end of the world to the other for which we use online servers and services like ThingSpeak, an IoT analytics platform service that allows you to aggregate, visualise, and analyse live data streams in the cloud. Our machine might be powerful enough to send data every second, but the ultimate throughput is decided by slow data path allowed by those online servers.

This project can help us send sensors' data very frequently and also free and independent of bandwidth of the severs that are provided by WebSocket. (WebSocket is a computer communications protocol, providing full-duplex communication channels over a single TCP connection.) We achieve this using a better, secure yet globally accessible freemium, cross-platform, cloud based instant messaging service—the Telegram Messenger. We can send data every couple of seconds using a low-cost ESP32 development board to a Telegram bot channel and then receive it on the other end of the world using a simple Python code.

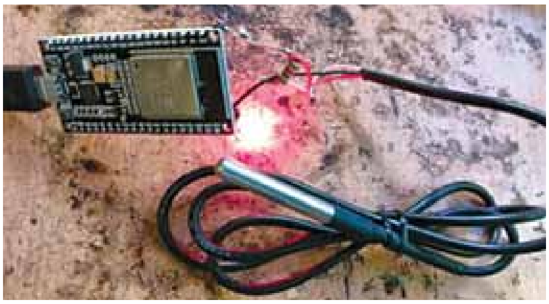

Fig. 1: Author's prototype

We have a remote telemetry station which sends data every four seconds. The data comes to the base station built around ESP32 board. The ESP32 is connected to internet Wi-Fi connection on the other end like a router. It re-sends data to a Telegram bot channel and at receiver end a normal computer connected to the internet runs Python3 to receive data continuously and put it in use. The working could be like this:

Field Sensors→To ESP32→To Telegram Channel→To Cyberspace→To far end Receiver→Wi-Fi Internet + Python→Reproduce data.

BILL OF MATERIAL		
ESP32	-	1
DS18B20	-	1
3.5mm LED	-	1
5.6k Resistor	-	1
USB Power supply	-	1

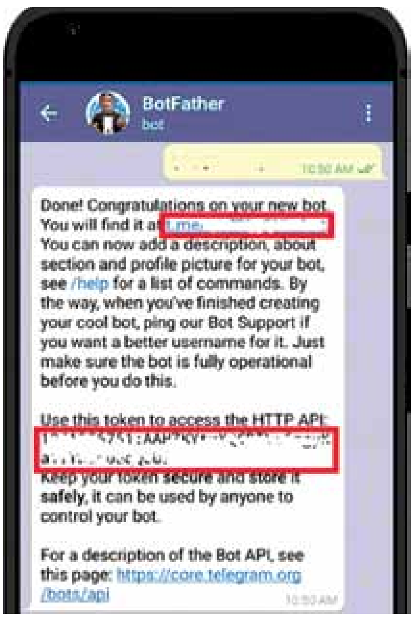

Fig. 2: Telegram bot API

The author's prototype is shown in Fig. 1 and its bill of material is shown in the table below.

Install the Telegram app in your mobile phone, tablet, or laptop. The app is available free at Google Play Store (for Android phones) and App Store (for iPhones), etc. Create a channel in Telegram to communicate with the ESP32 board.

After installing the app and setting up your account, search Botfather in the app. As soon as you open Botfather, you will see a Start or a Restart button. It will open a list of commands and their applications. Click on the /newbot command.

After this command, you need to give your bot a name. I named it bera_arduino. Next, set the username. While setting the username, keep in mind that it should be unique and it should end with the word bot, like bera1bot. As soon as you set the username, your bot is created, and you will see an API token. Save it somewhere as it will be needed later.

Fig. 2 shows the Telegram bot AI and Fig. 3 shows the Telegram bit setting.

Now, two parameters need to be found out for this (bera_arduino) channel—chat_id and bot_token—to prepare the Sketch software. After having created the chat name and user ID, give /mybots command; it will open your bots. Select your chosen bot and you will be presented with a graphical window. Press the API_token and you will receive the write bot_token for your bot. Image